

JAVA FRESHER MOST ASKED QUESTIONS (2025) – 100 FAQ WITH ANSWERS

1. What is Java?

Java is a high-level, object-oriented programming language used for building platform-independent applications.

2. What makes Java platform-independent?

Java uses bytecode executed by JVM, which works on any OS.

3. What is JDK?

Java Development Kit used to compile and run Java programs.

4. What is JRE?

Java Runtime Environment that contains JVM + libraries required to run Java.

5. What is JVM?

Java Virtual Machine that executes Java bytecode.

6. What is bytecode?

Intermediate code generated after compilation, executed by JVM.

7. What are OOP principles?

Encapsulation, Inheritance, Polymorphism, Abstraction.

8. What is Encapsulation?

Wrapping data & methods into a single unit (class).

9. What is Abstraction?

Hiding implementation details and showing only functionality.

10. What is Inheritance?

Acquiring properties of one class into another using 'extends'.

11. What is Polymorphism?

Ability of object to take many forms (overloading & overriding).

12. What is method overloading?

Same method name, different parameters.

13. What is method overriding?

Same method signature in subclass, runtime polymorphism.

14. What is constructor?

Special method to initialize objects.

15. Types of constructors?

Default, Parameterized, Copy constructor (manual).

16. What is 'this' keyword?

Refers to current object.

17. What is 'super' keyword?

Refers to parent class object.

18. What is an interface?

Blueprint with abstract methods.

19. What is an abstract class?

Class that cannot be instantiated and may contain abstract methods.

20. Abstract class vs interface?

Interface = full abstraction. Abstract class = partial abstraction.

21. What is a package?

Collection of related classes and interfaces.

22. What is a Java Bean?

Reusable component with private fields and getters/setters.

23. What is a static method?

Method belonging to class, not object.

24. What is final keyword?

Used to make variable/method/class non-modifiable.

25. What is String?

Immutable sequence of characters.

26. Why String is immutable?

Security, caching, efficient memory management.

27. StringBuilder vs StringBuffer?

StringBuilder is non-synchronized (faster). StringBuffer is synchronized.

28. Difference between == and equals()?

== compares references; equals() compares values.

29. What is wrapper class?

Object representation of primitive types.

30. Autoboxing?

Converting primitive to object automatically.

31. Unboxing?

Converting object to primitive.

32. What is exception?

Runtime error that disrupts normal flow.

33. Exception handling keywords?

try, catch, finally, throw, throws.

34. Checked vs unchecked exceptions?

Checked at compile time; unchecked at runtime.

35. What is ArrayList?

Resizable array implementation of List.

36. LinkedList?

Doubly linked list implementation.

37. HashSet?

Stores unique elements, no order.

38. TreeSet?

Stores elements in sorted order.

39. HashMap?

Stores key-value pairs, allows null.

40. Hashtable?

Synchronized version of HashMap.

41. What is vector?

Thread-safe dynamic array.

42. What is stack?

LIFO data structure.

43. What is queue?

FIFO data structure.

44. What is priority queue?

Queue sorted based on natural ordering.

45. What is iterator?

Object used to traverse collections.

46. What is fail-fast iterator?

Throws exception if structure modified.

47. What is fail-safe?

Does not throw error when modified.

48. What is String pool?

Memory area where String literals are stored.

49. What is multithreading?

Executing multiple threads simultaneously.

50. How to create thread?

Extend Thread or implement Runnable.

51. What is synchronization?

Control thread access to shared resource.

52. What is deadlock?

Two threads waiting for each other forever.

53. What is garbage collection?

Automatic memory cleanup.

54. What is JVM heap?

Memory for objects.

55. What is stack memory?

Stores method calls and local variables.

56. What is classloader?

Loads classes into JVM.

57. What is JDBC?

API for database connectivity.

58. Steps in JDBC?

Load driver, connect DB, execute query, close.

59. What is statement?

Used to execute SQL queries.

60. PreparedStatement?

Pre-compiled SQL, prevents SQL injection.

61. CallableStatement?

Used to call stored procedures.

62. What is servlet?

Java class handling HTTP requests.

63. What is doGet()?

Processes GET requests.

64. What is doPost()?

Processes POST requests.

65. What is session?

Server-side storage of user data.

66. What is cookie?

Client-side storage of small data.

67. What is JSP?

Java code inside HTML.

68. What are JSP directives?

page, include, taglib.

69. MVC architecture?

Model-View-Controller design pattern.

70. What is REST API?

Web service using HTTP methods.

71. What is JSON?

Lightweight data format.

72. What is Spring Boot?

Framework for fast Java development.

73. Features of Spring Boot?

Auto-config, embedded server, easy setup.

74. What is dependency injection?

Object is given instead of being created.

75. What is IoC container?

Controls object creation.

76. What is JPA?

Java Persistence API for ORM.

77. What is Hibernate?

ORM tool in Java.

78. What is entity?

Class mapped to DB table.

79. What is primary key?

Unique identifier.

80. What is fetch type?

Defines when related data loads.

81. What is microservice?

Small independent service.

82. Monolithic vs microservice?

Monolithic = large unified. Microservice = small independent services.

83. What is API gateway?

Entry point for microservices.

84. What is version control?

Tracks code changes.

85. What is Git?

Version control tool.

86. What is GitHub?

Cloud-based Git repository.

87. Git clone?

Download repo.

88. Git push?

Upload commits.

89. Git pull?

Fetch & merge.

90. CI/CD?

Continuous Integration / Continuous Deployment.

91. What is Docker?

Containerization platform.

92. What is container?

Lightweight isolated environment.

93. What is build tool?

Automates compilation (Maven/Gradle).

94. POM.xml?

Maven configuration file.

95. What is unit testing?

Testing individual components.

96. JUnit?

Java testing framework.

97. What is logging?

Recording events (Log4j/SLF4J).

98. What is API exception handling?

Returning meaningful error messages.

99. What is Swagger?

API documentation tool.

100. Why hire you as a Java fresher?

Because I have strong fundamentals, quick learning ability, and real project experience.